Runtime Visibility Limits of Bias Probes for Quantum Error-Mitigated Selection

A Preregistered Fixed-Budget Study

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Abstract

Confidence intervals quantify sampling uncertainty but do not, by themselves, bound differential systematic bias. A quantum decision can therefore be reported with high sampling confidence while selecting the wrong candidate. We formulate runtime bias diagnosis as a visibility problem: which probes can detect a bias large enough to reverse a QAOA–MaxCut top-1 decision, which are blind by construction, and when probe shots are worth removing from the target measurement budget. Protocol v4 preregisters six bias probes against seven non-null gate and readout structures (42 cells), with a fixed total budget of 16,384 shots per graph–cell–repeat. The confirmatory design uses 100 holdout graphs and 32 repeats; analytic visibility strata, matched nulls, operating rules, seeds, and decision endpoints were frozen before noisy development or holdout evaluation. The primary endpoint is the change in false-certification rate between a CI-only rule and a bias-aware abstaining rule at equal total cost. This pre-results manuscript reports only analytic results, split-99 software-design checks, a documented defect in the superseded v3 protocol, and a non-circular power calculation. No development or confirmatory holdout result is reported in this manuscript snapshot, and no hardware or physical-probe performance result is reported. The central contribution is thus a falsifiable and auditable test of whether runtime diagnostics add decision value rather than merely respond to noise.

1 Introduction

Quantum error mitigation (QEM) trades additional circuits, shots, or model assumptions for reduced estimator error, without implementing full fault tolerance. Canonical examples include zero-noise extrapolation and related near-term mitigation methods[10, 7, 4]. Their performance is commonly summarized by estimator bias, variance, or mean-squared error. Selection tasks impose a different requirement: only the sign of a contrast between candidates determines the action. Improved average accuracy need not imply a more reliable decision, particularly when the two candidates acquire different systematic biases.

Let the ideal candidate values be μ_A and μ_B, with ideal gap $g = \mu_A - \mu_B$ and margin $\Delta = |g|$. Under an error process, let the population gap be $g_{\mathrm{n}} = g + b_A - b_B$. The decision-relevant bias is therefore

$$\beta = (b_A - b_B)\,\mathrm{sign}(g). \tag{1}$$

The ideal decision is reversed whenever $\Delta + \beta < 0$. A narrow interval around g_{n} can exclude zero and still certify the wrong sign, because the interval controls finite-shot variation around a biased center.

This study asks whether runtime probes can reveal such harmful differential bias under a fixed shot budget. The question is deliberately narrower than general device characterization. A useful probe must satisfy three conditions: it must respond to the harmful direction, retain adequate finite-shot power, and intercept enough wrong certifications to compensate for the target shots it consumes. A probe that fails the first condition is blind in principle; a probe that fails the second is visible only at the population level; and a probe that fails the third can be diagnostically informative but harmful to the deployed decision rule.

The framework draws on partial identification, where observations may define a set rather than a point[6]; negative controls, whose validity requires a shared bias mechanism[8]; and known learnability limits for quantum noise[2]. It also connects to established gate, readout, and randomized-compiling tools [1, 9, 11, 3]. The novelty claim is not that any individual probe or concentration bound is new. It is the preregistered joint test of analytic visibility, finite budget, and decision-level false certification across a common QAOA selection task.

The study is simulation-based. No present claim establishes hardware utility, and the synthetic three-body readout stress structure has no asserted hardware provenance. Hardware experiments are a later go/no-go stage, conditional on the simulation results and independently frozen procedures.

2 Methodology

2.1 Decision, identification, and visibility

For QAOA–MaxCut[5], a bit string z on graph $G = (V, E)$ receives the normalized score

$$s_G(z) = \frac{1}{|E|} \sum_{(i,j)\in E} \mathbb{1}\{z_i \neq z_j\} \in [0, 1]. \tag{2}$$

Each graph has two frozen candidate parameter vectors. Their expected scores define g; the sign of an observed sample gap defines the selected candidate.

Let D denote all target and probe observations and $\mathcal{M}(D)$ the admissible error mechanisms compatible with them. A generic identification region for the ideal gap is

$$\mathcal{I}_g(D) = \{g(M) : M \in \mathcal{M}(D)\}. \tag{3}$$

If the region contains both positive and non-positive values, the decision is not sign-identified. This is a population concept, distinct from a confidence interval for shot noise. The project uses it to state blindness, not to claim that all 42 cells require explicit optimization of Eq. (3).

For each probe–structure cell, Protocol v4 freezes a component statistic and a matched null. A cell is *invariant* if the population statistic is identical under the structure and its matched null; no threshold can then recover mechanism-specific power. A *sensitive* cell has a nonzero population contrast, but may remain underpowered at the frozen budget. The visible/partial/blind label combines this analytic relation with the frozen power and false-alarm rules. Invariant and sensitive cells are reported separately so that construction-driven agreements cannot mask finite-budget failures.

2.2 Probes

Table 1 summarizes the six primary probes. P0, the target-only confidence-interval comparator, is intentionally excluded from the visibility matrix because an unresolved target interval is not a bias alarm. Two further statistics are recorded but are exploratory and carry no prediction: P1b, the negative quasi-probability weight of the scale-1/3 Richardson distribution built from P1 counts, and GVS, a covariance-survival ratio computed from target counts. Both reuse existing counts at no extra shot cost. Positive-cost probes receive $n_P = 1{,}638$ shots and the targets receive $n_T = 14{,}746$; zero-extra-shot P5 retains all $N = 16{,}384$ target shots.

Within each width–depth stratum and cell component, development matched-null scores $T_{(1)} \leq \cdots \leq T_{(n)}$ determine

$$\tau = T_{\min\{n,\lceil (n+1)(1-\alpha)\rceil\}}, \qquad \alpha = 0.05, \tag{4}$$

and the alarm is strict: $T > \tau$. Decision endpoints instead calibrate alarms against clean E0, reflecting deployment without knowledge of the active error structure. This separation prevents a structure-specific null from leaking oracle information into the operational rule.

2.3 Fixed-budget decision rules

R0 is the CI-only comparator. It certifies when the target gap interval excludes zero. R1 allocates 10% of the budget to the structure-designated probe and certifies only when the target interval excludes zero and the probe does not alarm. Let W_f be the event that the reduced-shot target decision is wrong and A_f the alarm event. Define

$$q_f = \mathbb{P}(W_f), \quad d_f = \mathbb{P}(A_f \cap W_f), \quad R_f = q_f - d_f. \tag{5}$$

Against an all-target error probability q_0, the probe lowers wrong-decision risk if and only if

$$d_f > q_f - q_0. \tag{6}$$

The left side is useful interception; the right side is the error cost of removing target shots. Equation (6) is exact for the abstaining policy and requires no independence assumption. Because abstention can mechanically reduce recorded errors, the study reports false certification and coverage separately rather than selecting a post hoc scalar loss.

3 Experiment Setup

3.1 Graphs, candidates, and error structures

The confirmatory holdout contains 100 independently generated graphs, four in each width–depth stratum for widths 4–8 and depths 1–5. Graph families are 34 Erdős–Rényi, 33 Watts–Strogatz, and 33 Barabási–Albert. Each graph has two frozen QAOA candidates and 32 scoring repeats per structure, probe, and parameter condition. Candidates are compiled to `rz,sx,x,cx` with identity layout, no routing, and optimization level zero. One candidate receives a logically cancelling CX–CX pair after each cost layer, creating a frozen gate count contrast without changing its ideal unitary.

Table 2 lists the seven non-null structures. E5 and E5′ retain their independent marginal flips in the matched null; E6 retains its baseline assignment rates. These matched nulls repair a v3 defect in which probes could alarm on the baseline construction rather than on the tested correlation or drift component.

E0 is a clean global null with equal deterministic offsets added to both candidates. It cannot change a decision contrast. E7, fake-backend snapshots, probe combinations, and additional diagnostics are exploratory and outside the 42-cell primary matrix.

3.2 Preregistered matrix and endpoints

The primary matrix comprises six probes by seven structures: 42 cells with 4 visible, 13 partial, and 25 blind predictions. Analytic classification yields 23 invariant and 19 sensitive cells (Table 3). The strata are never pooled into one headline agreement score. All 23 invariant cells

Table 1: Each probe targets one projection of the error process, and the preregistered matrix predicts it to be blind to most structures; no probe is treated as an omnibus detector. Positive-cost probes take $n_P = 1{,}638$ of the $N = 16{,}384$ shots; P5 reuses target counts. The last column lists the structures whose cells are predicted blind (Table 3).

Probe	Frozen statistic	Extra cost	Physical target	Predicted blind under
P1	$\lvert(g_3 - g_1)/2\rvert$ from local unitary folding at scales 1 and 3	2 circuits, n_P	Noise that scales with gate count	E2, E4, E5, E5′, E6
P2	$\lvert g_{\text{twin,obs}} - g_{\text{twin,ideal}}\rvert$ for a deterministic nearest-Clifford twin	2 circuits, n_P	Gate bias transferable to the twin	E5′
P3	$\lvert g_{\text{plain}} - \overline{g}_{\text{frame}}\rvert$ over 16 Pauli-frame versions per candidate	32 circuits, n_P	Coherent error around two-qubit gates	E1, E2, E4, E5, E5′, E6
P4a	Single-qubit assignment deviation from all-zero/all-one preparations; temporal contrast for E6	2 circuits, n_P	Readout marginals and drift	E1, E2, E3, E5, E5′
P4b	One frozen component per cell: marginal, pair-correlation, or temporal contrast from pair preparations	$4\binom{w}{2}$ circuits, n_P	Pairwise readout correlation	E1, E2, E3, E5′
P5	Bit-complement symmetry distance of target counts	none	Flip-symmetry-breaking error	E1, E2, E5, E5′

are predicted blind; the sensitive stratum contains every visible and partial prediction and two blind cells, P1/E6 and P3/E6, whose small block-dependent population contrast is predicted to stay below the finite-budget power boundary.

The primary estimand is

$$\Delta_{\text{FC}} = \frac{1}{7}\sum_{s=1}^{7}\{\text{FC}_s(R0) - \text{FC}_s(R1)\}, \tag{7}$$

where false certification means certifying the wrong ideal winner. A 95% interval is obtained from 10,000 stratified graph-cluster bootstrap resamples with shared graph indices across rules and structures. The prespecified outcome is "improvement" when the lower endpoint is positive, "no decision problem" when the 97.5th bootstrap percentile of pooled CI-only false certification is below 0.01, and "inconclusive" otherwise.

Secondary endpoints include coverage loss, interception among CI-only false certifications, behavior on flip and non-flip subsets, a structure-agnostic P2 rule, and visibility agreement within each analytic stratum. Readout certificates are restricted to cells whose probe measurements address the modeled readout component; they are not interpreted as certificates against gate or state-preparation noise.

3.3 Reproducibility and evidence separation

All stochastic artifacts derive from a single NumPy `SeedSequence` key containing split, artifact, structure, probe, width, depth, graph, candidate, repeat, and parameter indices. Split 10 is development; split 20 is holdout; split 99 is reserved for software tests and analytic design checks and is never reported as physical or confirmatory evidence. Raw and failed rows are append-only. Holdout access requires a second lock containing thresholds, operating points, manifest hashes, and platform policy.

Protocol v4 and its complete execution code were registered on OSF at https://osf.io/r63vn/ at 2026-11-13 01:45:49 UTC, commit `e3908dbdc9e73b3f5269195bf1839d74756e32e8`. It supersedes but does not erase the public v3 record (OSF `j67wy`, commit `e8b5300`). This manuscript snapshot uses repository HEAD `bdfeb5a`. At the snapshot date, the Task 2-2 development computation had been executed but its outputs were not yet validated or committed; none of them is opened, summarized, or used here.

4 Results

4.1 A0: analytic and noiseless design results

All results in this section are pre-data results: analytic derivations or split-99 software-design checks. They are not development, holdout, hardware, or physical-probe performance evidence.

QAOA–MaxCut has global bit-complement symmetry under the frozen ideal model, so $\langle Z_i \rangle = 0$. For uniform independent assignment noise with $\widetilde{Z}_i = \lambda Z_i + \delta$, each edge contrast scales by λ^2; hence

$$g_{\text{noisy}} = \lambda^2 g, \qquad \frac{\beta}{\Delta} = -(1 - \lambda^2) \in (-1, 0]. \tag{8}$$

Uniform E4-type readout noise therefore shrinks the gap but cannot reverse its sign in the population limit. The same bound applies to the frozen E6 mixture. This result

Table 2: Each non-null structure is paired with a matched null that keeps its baseline features and sets only the tested component to zero, so a probe cannot score visibility by reacting to the baseline. E5′ is a synthetic stress test, not a hardware-derived model.

Code	Physical interpretation	Frozen strength grid	Matched null
E1	Two-qubit depolarization after every hard CX	$p \in \{0.0025, 0.005, 0.01, 0.02, 0.04\}$	$p = 0$
E2	Every tagged cost rotation runs as $R_Z((1+\epsilon)\theta)$	$\epsilon \in \{\pm 0.005, \pm 0.01, \pm 0.02, \pm 0.04\}$	$\epsilon = 0$
E3	Coherent $R_{ZZ}(\zeta)$ immediately after every hard CX	$\zeta \in \{\pm 0.005, \pm 0.01, \pm 0.02, \pm 0.04\}$	$\zeta = 0$
E4	Independent asymmetric assignment error on every qubit	$(\Pr(1\|0), \Pr(0\|1))$: $(0.01, 0.02)$, $(0.02, 0.04)$, $(0.04, 0.08)$, $(0.08, 0.12)$, and the four swapped pairs	$(0, 0)$
E5	Correlated pair flips on the first edge, marginal rate 0.10	$\tau \in \{\pm 0.0025, \pm 0.005, \pm 0.0075, \pm 0.01\}$	$\tau = 0$, marginal flips kept
E5′	Synthetic three-body flips on the first connected triple, marginal rate 0.25	$\tau \in \{\pm 0.00390625, \ldots, \pm 0.015625\}$	$\tau = 0$, marginal flips kept
E6	Eight-block assignment drift, $e_0 = 0.04 + \delta t$, $e_1 = 0.06 - \delta t$	$\delta \in \{\pm 0.005, \pm 0.01, \pm 0.02, \pm 0.03\}$	$\delta = 0$, baseline kept

Table 3: The preregistered matrix predicts blindness for most cells, and every non-blind prediction lies in the budget-sensitive stratum. These are predictions, not observations.

Stratum	Visible	Partial	Blind	Total
Invariant	0	0	23	23
Sensitive	4	13	2	19
Total	4	13	25	42

limits what decision-flip evidence these structures can supply; it does not establish that their probes are ineffective for detecting readout change.

A noiseless density-matrix split-99 check over 50 graphs found no exact ideal ties. At the strongest selected signs, median β/Δ values were -0.7358 for E1, -0.0912 for E2, and -0.0781 for E3. The largest observed fraction with $\beta/\Delta \leq -1$ over the frozen gate grids was 0.12 (E2). These are deterministic design diagnostics used to select and stress the protocol; they are not estimates of holdout prevalence.

4.2 A1: the registered v3 defect was detectable before noisy data

After v3 registration and before any split-10 or split-20 row existed, an audit proved five definition-level conflicts. P4a/E5, P4a/E5′, P4b/E5′, P4a/E6, and P4b/E6 were predicted blind or partial, but their frozen statistics necessarily responded to baseline marginal or assignment error when calibrated against a perfectly clean E0. The corresponding visibility label would be visible by construction. Eight P0 cells were also identified as high risk, because target non-resolution is not a bias-visibility statistic; those eight were projections, not analytic proofs.

The audit motivated three surgical changes in v4: P0 became a comparator rather than a probe, every structure acquired a matched null, and each P4 cell received one frozen component rather than an omnibus maximum. The original v3 files and registration remain unchanged. This chronology is evidence of protocol repair, not evidence in favor of the v4 scientific predictions.

4.3 A2: semantic checks corrected four derivations

Before v4 registration, exact expected-count calculations evaluated every cell on 10 split-99 graphs of widths 4–6. The first pass disagreed with the analytic invariant/sensitive stratum in four cells: P3/E2, P5/E3, P1/E6, and P3/E6. Their derivations were corrected and both passes retained. A second pass over the same design had zero disagreements across 42 cells. Because it followed the corrections, this second pass is an internal consistency check, not an independent validation of the predictions.

4.4 A3: design power is limited by rare decision failures

Power analysis v5 used only split-99 exact-population inputs and a frozen grid of synthetic detection and false-alarm rates. It did not use visibility labels, development outputs, holdout outputs, device parameters, or hardware data. In the 50-graph design population, the strongest selected gate settings reversed 4 graphs for E1, 5 for E2, and 3 for E3; E4–E6 and E5′ reversed none. The CI-only false-certification mean was 0.0200 for E2 and zero for the other six structures at the modeled budget.

Table 4: The manuscript intentionally stops at pre-data evidence. Development, holdout, and hardware claims remain unavailable at this stage.

Evidence class	Status at snapshot
Analytic theory	Available; assumptions stated
Split-99 checks	Available; design/software evidence only
OSF protocol v4	Registered with execution code
Split-10 development	Executed; not yet validated or committed; excluded
Split-20 holdout	Not accessed
Hardware data	None

Across the preregistered design scenarios, the probability that the primary analysis would declare "no decision problem" was approximately 0.685–0.775, and the probability of an "improvement" conclusion did not exceed 0.15. The calculation therefore predicts that the main limitation may be the scarcity of false certifications to intercept, not merely probe sensitivity. This is a prospective design warning, not a study result.

5 Conclusion

Protocol v4 turns a broad question—whether a runtime diagnostic "sees" noise—into a fixed-budget decision test. It separates analytic blindness from finite-shot failure, uses matched nulls, and evaluates whether alarms intercept wrong certifications rather than merely reacting to an error channel. Before any reported development or holdout result, the project has established an auditable 42-cell prediction matrix, repaired a registered protocol defect, and shown prospectively that rare decision flips may dominate statistical power. It has not established that any probe improves decisions, generalizes to hardware, or detects unknown noise. Those conclusions remain contingent on the locked development and confirmatory stages.

6 Discussion

6.1 What the present evidence supports

The analytic results support three bounded claims. First, sampling confidence and differential-bias control are distinct. Second, probe visibility is mechanism-specific: a statistic can be exactly invariant to a harmful direction even when another probe responds. Third, a responsive probe is not necessarily worth its shots; decision value requires Eq. (6), not high alarm power alone. The documented v3 failure also shows why these claims require matched nulls and a strict separation between comparator uncertainty and bias alarms.

6.2 Hypotheses and possible explanations

The following are expectations to be tested, not findings. If the primary endpoint returns "no decision problem", the most direct explanation is the design calculation of Section 4.4: at this budget, CI-only false certification is concentrated in a few graphs under one structure, leaving little for any probe to intercept. If sensitive cells predicted partial are observed blind, a plausible explanation is that the population contrast is real but smaller than the frozen power boundary allows at $n_P = 1{,}638$. Disagreements in the invariant stratum would instead indicate an implementation or derivation error, because their population statistics are equal by construction.

6.3 Limitations

The error classes are structured simulations, not a representative sample of real devices. E5′ is synthetic and has no claimed hardware evidence. E4 and the frozen E6 cannot reverse the population decision under the idealized symmetry assumptions of Eq. (8); they test visibility and calibration, not a rich source of false certifications. Candidate generation, graph families, QAOA depths, and the deliberate CX–CX contrast limit external generalization. Partial-identification language does not remove assumptions about the admissible mechanism class. Moreover, semantic tests performed on the same design after corrections are not independent replication. The matched nulls are defined inside the simulated error model, so visibility labels depend on that model. The individual lemmas rest largely on established statistics and information theory; the contribution is their joint, preregistered use, not new bounds. Hardware validation has not been authorized under the current protocol.

The primary endpoint may be under-informative if false certifications are rare. In that case, a "no decision problem" outcome is scientifically different from evidence that probes work. Conversely, an inconclusive result cannot be converted post hoc into a positive visibility claim. Coverage and abstention cost must accompany every reduction in false certification.

6.4 Reproducibility and next evidence gates

The confirmatory lock must be written before split-20 access and must freeze the development-selected operating points, thresholds, hashes, and execution platform. Public continuous integration should reproduce the test suite and the deterministic reference phase on clean Windows and Linux runners, recording package and BLAS configurations. Platform differences should be retained and reported rather than hidden behind relaxed tolerances. Only after these gates and the holdout analysis should a hardware micro-study be considered. Such a study would be an external anchor, not retroactive validation of the simulated error family.

References

[1] S. Bravyi, S. Sheldon, A. Kandala, D. C. Mckay, and J. M. Gambetta. Mitigating measurement errors in multiqubit experiments. *Physical Review A*, 103(4):042605, 2021.

[2] S. Chen, Y. Liu, M. Otten, A. Seif, B. Fefferman, and L. Jiang. The learnability of pauli noise. *Nature Communications*, 14, 2023.

[3] P. Czarnik, A. Arrasmith, P. J. Coles, and L. Cincio. Error mitigation with Clifford quantum-circuit data. *Quantum*, 5:592, 2021.

[4] S. Endo, S. C. Benjamin, and Y. Li. Practical quantum error mitigation for near-future applications. *Physical Review X*, 8:031027, 2018.

[5] E. Farhi, J. Goldstone, and S. Gutmann. A quantum approximate optimization algorithm, 2014.

[6] G. W. Imbens and C. F. Manski. Confidence intervals for partially identified parameters. *Econometrica*, 72:1845–1857, 2004.

[7] Y. Li and S. C. Benjamin. Efficient variational quantum simulator incorporating active error minimization. *Physical Review X*, 7(2):021050, 2017.

[8] M. Lipsitch, E. Tchetgen Tchetgen, and T. Cohen. Negative controls: A tool for detecting confounding and bias in observational studies. *Epidemiology*, 21:383–388, 2010.

[9] F. B. Maciejewski, F. Baccari, Z. Zimborás, and M. Oszmaniec. Modeling and mitigation of cross-talk effects in readout noise with applications to the quantum approximate optimization algorithm. *Quantum*, 5:464, 2021.

[10] K. Temme, S. Bravyi, and J. M. Gambetta. Error mitigation for short-depth quantum circuits. *Physical Review Letters*, 119(18):180509, 2017.

[11] J. J. Wallman and J. Emerson. Noise tailoring for scalable quantum computation via randomized compiling. *Physical Review A*, 94:052325, 2016.

A Blindness lemma

Let M denote an admissible noise mechanism, Q_M the joint population law of all target and probe observations, and $d(M) \in \{-1, +1\}$ the ideal decision. If two admissible mechanisms M_+ and M_- satisfy

$$Q_{M_+} = Q_{M_-}, \qquad d(M_+) \neq d(M_-), \tag{9}$$

then no rule measurable with respect to the recorded observations can be uniformly correct over the model class. Indeed, any rule has the same output distribution under the identical laws, so its error probabilities under the opposite decisions sum to one. At least one is therefore at least $1/2$. This is an identification statement, not a finite-sample approximation.

B Finite-shot necessary condition

For two data-generating states P_0 and P_1 that require opposite binary decisions, any test based on N independent observations obeys

$$P_0(\text{error}) + P_1(\text{error}) \geq 1 - \mathrm{TV}(P_0^N, P_1^N). \tag{10}$$

Using $\mathrm{TV}(P, Q) \leq H(P, Q)$ and $H^2(P_0^N, P_1^N) \leq N H^2(P_0, P_1)$ under the convention $H^2(P, Q) = \int (\sqrt{p} - \sqrt{q})^2$, a necessary condition for both errors to be at most $\alpha < 1/2$ is

$$N\, H^2(P_0, P_1) \geq (1 - 2\alpha)^2. \tag{11}$$

The bound is pairwise and necessary; it neither constructs an optimal probe nor replaces the exact protocol calibration.

C Local information is not global sign identification

Two analytic counterexamples separate a local Fisher null-space condition from global sign identification. For $X \mid \theta \sim \mathcal{N}(\theta^2, 1)$ and $g(\theta) = \theta$, Fisher information is positive at every $\theta \neq 0$, yet $P_a = P_{-a}$ and the two equivalent points have opposite decisions. Conversely, for $X \mid (u, v) \sim \mathcal{N}(u, 1)$ with $v \in (-1/2, 1/2)$ and $g_m(u, v) = m(1 + v)$, the unobserved v direction changes the gap magnitude and violates the local null-space condition, but the full identification region $(m/2, 3m/2)$ remains positive. Local sensitivity and global decision blindness therefore do not imply one another without additional regularity and localization assumptions.

D Fixed-budget Proposition 2

Fix a total budget N, give a fraction $f \in [0, 1)$ to the probe and $(1 - f)N$ to the target, and return *unresolved* whenever the probe alarms. With W_f, A_f, q_f, and d_f as in Section 2.3, the wrong-decision probability is exactly

$$R_f = \mathbb{P}(W_f \cap A_f^c) = q_f - d_f, \tag{12}$$

and the all-target baseline is $R_0 = q_0$. Hence $R_f < R_0$ if and only if $d_f > q_f - q_0$, which is Eq. (6). Writing $\eta_f = \mathbb{P}(A_f \mid W_f)$ for $q_f > 0$ gives the equivalent condition

$$\eta_f > 1 - q_0/q_f. \tag{13}$$

No Gaussian or independence assumption is used. Under an additional Gaussian model with independent probe and target data, $q_f = \Phi(-m\sqrt{(1 - f)N}/\sigma)$ for noisy gap $m = \Delta + \beta > 0$, and η_f equals the probe's alarm probability;

this approximation does not apply to P5, P1b, or GVS, which reuse target counts. If unresolved outcomes carry loss λ, the probe lowers $R_f + \lambda\mathbb{P}(A_f)$ if and only if $d_f > q_f - q_0 + \lambda\mathbb{P}(A_f)$. The study reports false certification and coverage separately instead of fixing λ post hoc. When $m < 0$, removing target shots moves q_f toward 1/2 and can lower q_f mechanically; that reduction is not evidence of visibility.

E Protocol correction chronology

Table 5: Protocol changes were made before split-10 or split-20 evidence was used. Earlier registrations remain public rather than being overwritten.

Stage	Public record and scientific role
v2 review	Pre-development review proved that uniform readout shrinks but does not reverse the frozen QAOA decision; it motivated structure-specific operating points.
v3	OSF `j67wy`, commit `e8b5300`; retained unchanged after a pre-data audit found P0-role and null-calibration defects.
v4	OSF `r63vn`, commit `e3908db`; registered matched nulls, 42-cell strata, the decision endpoint, and complete execution code.
Snapshot	Repository `bdfeb5a`; Task 2-2 uncommitted development outputs are excluded from this manuscript.